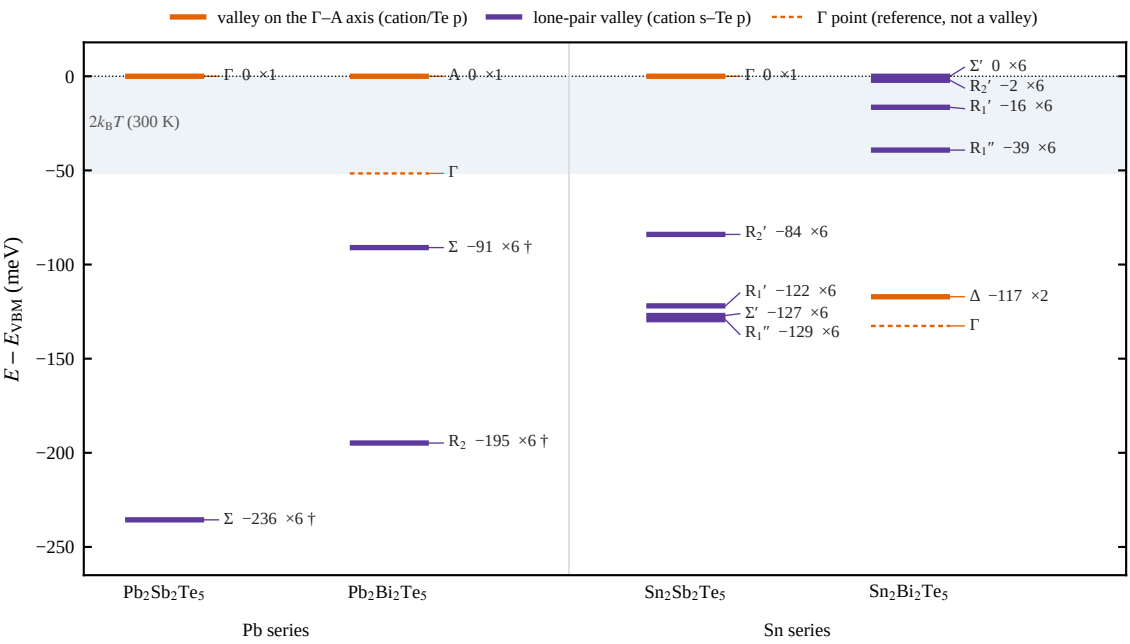
valley on the Γ–A axis (cation/Te p)
lone-pair valley (cation s–Te p)
Γ point (reference, not a valley)
$E - E_{VBM}$ (meV)
0
−50
−100
−150
−200
−250
$2k_BT$ (300 K)
Γ 0 ×1
A 0 ×1
Γ 0 ×1
Σ′ 0 ×6
R2′ −2 ×6
R1′ −16 ×6
R1″ −39 ×6
Γ
Σ −91 ×6 †
R2′ −84 ×6
R1′ −122 ×6
Σ′ −127 ×6
R1″ −129 ×6
Δ −117 ×2
Γ
R2 −195 ×6 †
Σ −236 ×6 †
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
Pb series
Sn series